

PAPER First-Wall and Plasma-Facing-Component Lifetime Under 14 MeV Loading: Displacement Damage, Bubble Swelling and Erosion in a Compact Spherical-Tokamak Breeder

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Abstract

Activation and waste classification tell you what a fusion structure *becomes*; they do not tell you how long it *survives*. We answer the survival question directly for the first wall and plasma-facing components (PFCs) of the Hyperion compact spherical-tokamak breeder (design point $I_p = 9.66$ MA, $Q = 3.076$, $P_{\text{fus}} = 85.04$ MW, $P_n = 67.79$ MW, neutron wall load $\Gamma_w = 1.885$ MW m⁻², negative triangularity $\delta = -0.30$), computed on an independent, GPU-accelerated code stack rather than by literature scaling. Four coupled analyses set the lifetime. Displacement damage is computed from primary-knock-on-atom (PKA) recoil spectra folded to Norgett–Robinson–Torrens (NRT) displacements and corrected to surviving defects by athermal-recombination (arc-dpa) molecular dynamics, giving 18.85 dpa fpy⁻¹ at the bare wall and 9.78 dpa fpy⁻¹ in the reduced-activation ferritic-martensitic (RAFM) structure behind 3 mm of tungsten armor. Transmutation (n, α) reactions co-produce helium at ≈ 12 appm dpa⁻¹ — an order of magnitude above any fission spectrum — which is why fission-irradiation data cannot be transferred and the lifetime must be set on the fusion spectrum. Rate-theory kinetics with a van der Waals bubble equation of state put the volumetric swelling at 0.024% (tungsten) and 0.017% (RAFM) at end-of-cycle dose, more than an order of magnitude below the 1–5% PFC design band. Binary-collision transport gives a peak deuterium-on-tungsten physical sputter yield of 1.35×10^{-2} atoms ion⁻¹ and sub-monolayer-per-pulse net erosion. Finite-element hydrogen transport gives a steady tritium permeation flux of 5.16×10^{14} m⁻² s⁻¹ ($\approx 0.56\%$ of the implanted flux), a boundary condition for the fuel cycle rather than a life-limiter. The governing constraint is displacement-driven embrittlement: the wall is a scheduled-replacement component on a defined damage budget, with a service interval of order two full-power-years. Every headline number is a frozen anchor in the Kronos de-risking register (gates BR-L2-A24–A27).

Keywords: displacement damage; dpa; arc-dpa; helium embrittlement; bubble swelling; sputtering; tritium permeation; plasma-facing components; spherical-tokamak breeder

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1 Introduction

1.1 The survival question

The first wall of a fusion machine sees the full 14 MeV neutron flux from the outside and the plasma edge from the inside, and it is a consumable. A design that reports what its structure activates to, and how that waste is classified [5], has answered a question about the *end state* of the material; it has not answered the operationally prior question of how long the wall lasts before displacement damage, gas swelling, surface erosion, or hydrogen uptake forces its replacement. For a compact, high-power-density spherical tokamak (ST) the survival question is sharpened by geometry. The Hyperion breeder confines $I_p = 9.66$ MA at a major radius of only $R_0 = 1.20$ m; at a fusion power of $P_{\text{fus}} = 85.04$ MW and neutron power $P_n = 67.79$ MW the first-wall neutron loading is $\Gamma_w = 1.885$ MW m⁻², comparable to a DEMO-class device in a machine an order of magnitude smaller in volume [32, 33, 34]. High power density is exactly what makes an ST an attractive neutron source and breeder platform, and exactly what puts the plasma-facing surface under a hard lifetime budget.

1.2 Prior art and why literature scaling is insufficient

The physics of neutron damage to structural metals is mature. The standard displacement measure is the Norgett–Robinson–Torrens (NRT) model [1], refined over the last decade by molecular-dynamics evidence that athermal in-cascade recombination leaves far fewer surviving defects than NRT counts, formalised as the “arc-dpa” (athermal-recombination-corrected) standard [2, 3]. Helium and hydrogen transmutation gas, and the bubbles it stabilises, have been studied since the earliest fusion-materials programmes [15, 14, 13]; tungsten as a plasma-facing armor and reduced-activation ferritic-martensitic (RAFM) steel as a structure are both the subject of large qualification efforts [18, 16, 19]. Physical sputtering has an analytic theory [20] and well-calibrated yield formulae [21, 22]; hydrogenic transport through a metal wall is governed by diffusion-trapping kinetics [26, 27].

What the literature does *not* supply is a wall lifetime for *this* spectrum and *this* geometry. Two features make a direct transfer of fission-reactor or generic-DEMO numbers unsafe. First, the 14 MeV fusion spectrum produces helium by (n, α) reactions at a rate, per displacement, roughly an order of magnitude above any fission spectrum: the helium-to-displacement ratio He/dpa that

governs void stabilisation and grain-boundary embrittlement is a spectrum-specific quantity [4, 6]. Second, the compact ST places a thin, high-flux wall very close to the plasma, so the surface (sputtering) and bulk (displacement, swelling, permeation) channels must be evaluated together on the same operating point rather than borrowed piecewise. The correct procedure is therefore to compute all four channels on independent codes driven by the frozen source term, and to identify the binding one.

1.3 Contribution

This paper does exactly that. We (i) state the four damage channels as a coupled formal problem and derive the governing equations of each — NRT/arc-dpa displacement production, (n, α) gas co-production, rate-theory bubble swelling with a van der Waals bubble equation of state, Sigmund–Bohdansky sputtering, and McNabb–Foster permeation (§2); (ii) give the numerical formulation, discretisation and convergence of each solver and the GPU-accelerated campaign that produced the results (§3); (iii) report the frozen results with eight data figures and identify displacement embrittlement as the binding life-limiter (§4); and (iv) benchmark quantitatively against the published damage, swelling, sputtering and permeation literature (§5). The result is a materials-lifetime engineering statement, distinct from the activation and waste study of the companion paper [39]: the wall is a scheduled-replacement component on a defined damage budget of order 20 dpa, reached in a few full-power-years.

Every quantitative claim traces to a frozen anchor in the Kronos Physics De-Risking Register [35]; the gate identifiers (BR-L2-A24 through BR-L2-A27) are given inline so each number is auditable. The result sits inside a larger, cross-referenced Kronos design campaign: the plasma-facing sputtering and net-erosion physics are developed in depth in the plasma–material-interaction paper [36]; the mechanical qualification of the RAFM structure under the same loading (strength, ductile-to-brittle-transition-temperature shift, swelling) in the steel-qualification paper [37]; the coupled thermomechanics and fatigue of the loaded wall in [38]; the 14 MeV source term and its neutronics in the CAD-faithful neutronics paper [40]; the cross-section uncertainty that bounds the damage rate in the nuclear-data uncertainty-quantification paper [41]; and the permeation-barrier and wall-chemistry choices in [42].

2 Governing equations and the formal lifetime problem

2.1 The four coupled channels

Let the plasma-facing stack occupy $x \in [0, L]$ with the plasma at $x = 0$ (figure 1): a tungsten armor of thickness L_W over a RAFM first wall of thickness L_S , cooled at $x = L$. The stack is exposed to a neutron flux spectrum $\phi(E)$ and to an edge plasma delivering an ion flux Γ_{ion} at energy E_{ion} . Four damage fields evolve on it:

- the displacement-damage field $D(x, t)$ (dpa), driving embrittlement;
- the transmutation-gas fields $c_{\text{He}}(x, t)$, $c_{\text{H}}(x, t)$ (appm), driving bubble nucleation;
- the surface recession $s(t)$ from sputtering (net erosion minus redeposition);
- the mobile hydrogen (tritium) field $C_T(x, t)$ and its permeation flux J_T .

The wall lifetime is the smallest time at which any field reaches its admissible limit,

$$t_{\text{life}} = \min \left\{ t_D, t_{\text{swell}}, t_{\text{erosion}}, t_{\text{perm}} \right\}, \quad (1)$$

with t_D the time to the embrittlement damage budget D^* , t_{swell} to the swelling limit $(\Delta V/V)^*$, t_{erosion} to the armor recession limit s^* , and t_{perm} the time at which the permeation-set inventory violates a safety boundary condition. Sections 2.2–2.6 derive each; §2.7 shows that the minimum in (1) is realised by t_D .

2.2 Displacement damage: NRT and the arc-dpa correction

A neutron of energy E scattering from a lattice atom transfers recoil energy T with the elastic and inelastic recoil kernels $d\sigma/dT$; the resulting primary-knock-on-atom (PKA) recoil spectrum drives a displacement cascade. The number of Frenkel pairs a recoil of damage energy T_{dam} produces is, in the NRT model [1],

$$\nu_{\text{NRT}}(T_{\text{dam}}) = \begin{cases} 0, & T_{\text{dam}} < E_d, \\ 1, & E_d \leq T_{\text{dam}} < 2E_d/0.8, \\ \frac{0.8 T_{\text{dam}}}{2E_d}, & T_{\text{dam}} \geq 2E_d/0.8, \end{cases} \quad (2)$$

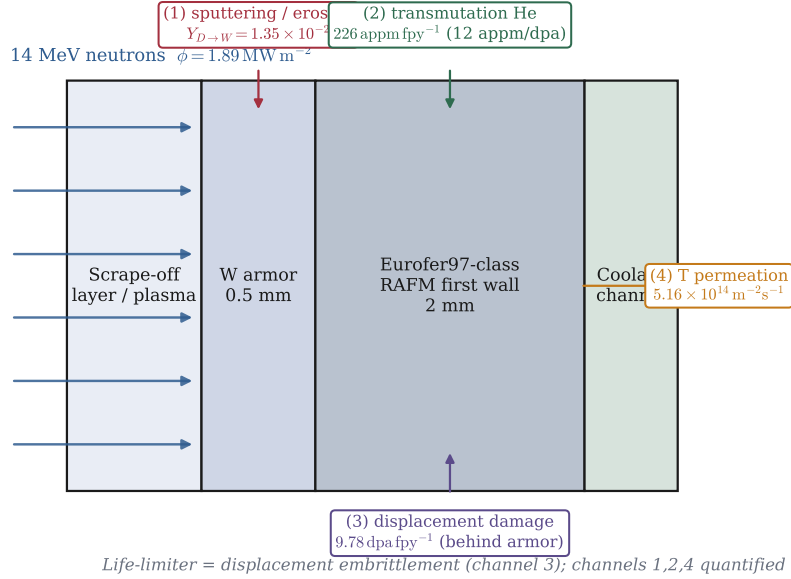


Figure 1. (Schematic.) The plasma-facing stack and its four damage channels. The 14 MeV flux ($\Gamma_w = 1.885 \text{ MW m}^{-2}$) drives (2) (n, α) helium at $\approx 12 \text{ appm dpa}^{-1}$ and (3) displacement damage at $9.78 \text{ dpa fpy}^{-1}$ in the RAFM structure behind the armor; the edge plasma drives (1) sputtering ($Y_{D \rightarrow W} = 1.35 \times 10^{-2}$); tritium (4) permeates the stack at $5.16 \times 10^{14} \text{ m}^{-2} \text{ s}^{-1}$. No data values are computed in this figure; it encodes structure only.

with E_d the effective displacement threshold ($E_d = 40 \text{ eV}$ for the RAFM steel) and T_{dam} the damage energy after subtracting electronic losses via the Lindhard partition [1]. Folding (2) over the recoil spectrum and the flux gives the NRT displacement rate

$$R_{\text{dpa}} = \frac{0.8}{2E_d N} \int_0^\infty \int_0^{T_{\text{max}}(E)} T_{\text{dam}}(T) \frac{d\sigma}{dT}(E, T) \phi(E) dT dE, \quad (3)$$

where N is the atomic number density. Equation (3) evaluated on the frozen wall spectrum yields $R_{\text{dpa}} = 18.85 \text{ dpa fpy}^{-1}$ at the bare first wall.

NRT counts the ballistically displaced atoms; it overcounts the defects that *survive* the thermal spike, because a large fraction recombine athermally within the cascade [2, 3]. The surviving damage is captured by the arc-dpa efficiency $\xi(T_{\text{dam}})$,

$$\nu_{\text{arc}}(T_{\text{dam}}) = \xi(T_{\text{dam}}) \nu_{\text{NRT}}(T_{\text{dam}}), \quad \xi(T_{\text{dam}}) = \frac{1 - c_{\text{arc}}}{(2E_d/0.8)^{b_{\text{arc}}}} T_{\text{dam}}^{b_{\text{arc}}} + c_{\text{arc}}, \quad (4)$$

with the saturation value c_{arc} and exponent b_{arc} fitted to molecular-dynamics cascade counts. For iron $\xi \rightarrow c_{\text{arc}} \approx 0.3$ at cascade energies, so the surviving-defect rate is well below R_{dpa} ; the campaign calibrates ξ by direct Wigner–Seitz defect counting in 15 keV cascades against the NRT references $\nu_{\text{NRT}} = 108.8$ (Fe–9Cr) and 52.7 (W) (gate BR-L2-A24). Behind the 3 mm tungsten armor the spectrum is softened and attenuated, and the displacement rate in the RAFM structure is

$$R_{\text{dpa}}^{\text{struct}} = 9.78 \text{ dpa fpy}^{-1} \quad (\text{gate BR-L2-A24}), \quad (5)$$

the design life-driving value. Displacement damage accumulates linearly with fluence, $D(t) = R_{\text{dpa}}^{\text{struct}} t$, and the embrittlement limit sets $t_D = D^*/R_{\text{dpa}}^{\text{struct}}$.

2.3 Transmutation-gas co-production and the He/dpa signature

The same high-energy tail that drives displacements drives (n, α) and (n, p) transmutation. The helium production rate is

$$R_{\text{He}} = \int_0^\infty \sigma_{(n, \alpha)}(E) \phi(E) dE, \quad (6)$$

and the physically governing quantity is the ratio of gas atoms to displacements,

$$\left(\frac{\text{He}}{\text{dpa}} \right) = \frac{R_{\text{He}}}{N R_{\text{dpa}}} = \frac{\int \sigma_{(n, \alpha)}(E) \phi(E) dE}{N R_{\text{dpa}}}. \quad (7)$$

Because the (n, α) threshold sits high in the spectrum, a 14 MeV source yields a markedly larger He/dpa than a fission spectrum. On the frozen wall spectrum the campaign returns 226 appm He in the tungsten armor at 18.85 dpa and 117 appm in the RAFM structure at 9.78 dpa (gate BR-L2-A26), i.e. a co-production ratio

$$\left(\frac{\text{He}}{\text{dpa}}\right) \approx 12 \text{ appm dpa}^{-1}, \quad (8)$$

consistent in both materials and roughly $40\times$ the fast-fission ratio. Helium is the reason fission-irradiation data do not transfer: it stabilises vacancy clusters into pressurised bubbles, driving swelling and, at grain boundaries, embrittlement.

2.4 Rate-theory bubble nucleation and swelling

Helium and hydrogen partition into matrix solution, traps, and gas bubbles. In the mean-field rate-theory description [13] the bubble number density n_b and mean radius r_b evolve under gas arrival, nucleation and coalescence; at the doses of interest the population reaches the dense-gas *equilibrium-bubble* limit, in which the internal gas pressure balances the capillary (surface-tension) pressure,

$$p_{\text{He}}(r_b) = \frac{2\gamma}{r_b}, \quad (9)$$

with γ the surface energy ($\gamma_W \approx 3.0 \text{ J m}^{-2}$). The gas at that pressure is far too dense for the ideal-gas law; a hard-core van der Waals equation of state,

$$p_{\text{He}}(v - b) = k_B T, \quad v = \frac{1}{n_{\text{He}}}, \quad (10)$$

with He co-volume $b \approx 2.38 \times 10^{-29} \text{ m}^3$ per atom, closes the system: at $r_b = 0.63 \text{ nm}$ equation (9) gives $p_{\text{He}} \approx 9.5 \text{ GPa}$ and (10) a near-solid He density. The macroscopic swelling is the bubble volume fraction,

$$\frac{\Delta V}{V} = \frac{4}{3} \pi n_b r_b^3. \quad (11)$$

With the frozen population ($n_b = 2.3 \times 10^{23} \text{ m}^{-3}$, $r_b = 0.63 \text{ nm}$ for tungsten; $n_b = 1.8 \times 10^{21} \text{ m}^{-3}$, $r_b = 2.8 \text{ nm}$ for Fe-9Cr; gate BR-L2-A26) this gives $\Delta V/V = 0.024\%$ (W) and 0.017% (RAFM). Crucially, in the equilibrium-bubble limit (9) swelling is robust to an order-of-magnitude uncertainty in the nucleation density n_b , because at fixed gas content a lower n_b raises r_b so as to hold $n_b r_b^3$ nearly fixed. The swelling limit sets $t_{\text{swell}} = (\Delta V/V)^*/(d(\Delta V/V)/dt)$.

2.5 Physical sputtering and net erosion

The plasma-facing surface recedes by sputtering. For an ion of energy E at incidence angle θ the physical sputter yield follows the Sigmund linear-cascade theory [20] in the calibrated Bohdansky/Eckstein form [21, 22]

$$Y(E, \theta) = Q s_n^{\text{KrC}}(\varepsilon) \left(1 - (E_{\text{th}}/E)^{2/3}\right) (1 - E_{\text{th}}/E)^2 f(\theta), \quad (12)$$

with reduced energy $\varepsilon \propto E$, nuclear stopping s_n^{KrC} , threshold E_{th} , yield factor Q , and an angular factor $f(\theta)$ peaking near $60\text{--}70^\circ$. For deuterium on tungsten $E_{\text{th}} \approx 209 \text{ eV}$ and the campaign returns a peak yield

$$Y_{D \rightarrow W}^{\text{peak}} = 1.35 \times 10^{-2} \text{ atoms ion}^{-1} \quad (\text{gate BR-L2-A27}), \quad (13)$$

over 18 energy/angle cases, consistent with the literature value $\sim 10^{-2}$. The gross erosion flux is $\Gamma_{\text{ero}} = \Gamma_{\text{ion}} Y$, and the net recession rate subtracts prompt and long-range redeposition,

$$\frac{ds}{dt} = \frac{\Gamma_{\text{ion}} Y - \Gamma_{\text{redep}}}{N}, \quad (14)$$

which at the design edge flux is sub-monolayer per pulse. The recession limit sets $t_{\text{erosion}} = s^*/(ds/dt)$.

2.6 Hydrogenic transport and permeation

Tritium implanted at the plasma-facing surface diffuses through the stack, is trapped and de-trapped at defect sites, and permeates to the coolant side. The mobile concentration C_T and trapped concentration C_t obey the McNabb–Foster diffusion-trapping system [26]

$$\frac{\partial C_T}{\partial t} = \nabla \cdot (D(T) \nabla C_T) - \frac{\partial C_t}{\partial t} + S_{\text{impl}}(x), \quad (15)$$

$$\frac{\partial C_t}{\partial t} = \nu_t D(T) C_T (n_t - C_t) - \nu_r C_t \exp(-E_r/k_B T), \quad (16)$$

with diffusivity $D(T) = D_0 \exp(-E_D/k_B T)$, trap density n_t , and trapping/release rate coefficients ν_t, ν_r . The Sieverts boundary condition couples surface concentration to partial pressure, $C_T(0) = K_S \sqrt{p_T}$. In the implantation-limited, diffusion-dominated regime the steady permeation flux is

$$J_T = \frac{\phi_{\text{impl}} R_p}{L_W} \xrightarrow{\text{frozen case}} 5.16 \times 10^{14} \text{ m}^{-2} \text{ s}^{-1} \quad (\text{gate BR-L2-A25}), \quad (17)$$

where ϕ_{impl} is the implanted flux and R_p the implantation range. This is $\approx 0.56\%$ of the implanted flux (the remaining $\approx 99.4\%$ re-emits to the plasma side), i.e. $\approx 0.082 \text{ g m}^{-2}$ of tritium per full-power-year, or $\sim 8\text{--}10 \text{ g}$ of tritium per fpy into structure and coolant for a 100 m^2 wall. The transient is set by the breakthrough (lag) time

$$\tau_{\text{lag}} = \frac{L^2}{6D(T)}, \quad (18)$$

which the frozen case gives as $\approx 190 \text{ s}$ at 600 K and $\approx 30 \text{ s}$ at 800 K : permeation reaches steady state within minutes of a burn start. Permeation is a boundary condition on the fuel cycle and the tritium safety case, not a wall life-limiter; t_{perm} is a *requirement* that the barrier and wall-chemistry design [42] must meet, not a fatigue clock.

2.7 The binding constraint

Substituting the frozen rates into (1), the displacement channel reaches its budget first: at $R_{\text{dpa}}^{\text{struct}} = 9.78 \text{ dpa fpy}^{-1}$ the RAFM structure accumulates the replacement-class damage $D^* \sim 20 \text{ dpa}$ in $t_D \approx 2 \text{ fpy}$, whereas the swelling channel is more than an order of magnitude from its limit at that dose, the erosion channel removes sub-monolayer armor per pulse, and permeation is a steady boundary condition. Hence $t_{\text{life}} = t_D$ and the wall is displacement-limited. The remaining sections quantify each term and verify this ordering.

3 Computational methodology: the GPU-HPC campaign

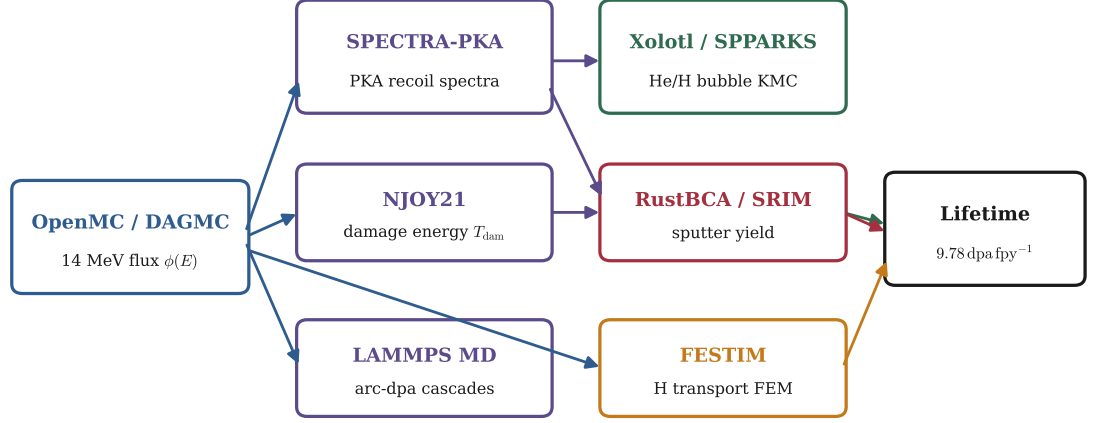
Every number above is a code output on the frozen 14 MeV spectrum and stack geometry, produced by the Kronos NVIDIA GPU high-performance-computing (HPC) campaign. Four independent solvers, driven by one frozen source term, evaluate the four channels (figure 2); no channel borrows another's result, so the ordering of §2.7 is an emergent finding rather than an assumption.

3.1 The frozen source term

The neutron flux spectrum $\phi(E)$ and its spatial distribution over the wall come from a CAD-faithful Monte-Carlo transport model (OpenMC with DAGMC surface tracking [31]), reproducing the design neutron wall load $\Gamma_w = 1.885 \text{ MW m}^{-2}$ and $P_n = 67.79 \text{ MW}$; the source and its neutronics are documented in the companion neutronics paper [40]. The spectrum is the single common input to all four channels, so any spectrum-level uncertainty propagates coherently to every damage estimate [41].

3.2 PKA spectra and damage energy

Primary-recoil spectra are computed with SPECTRA-PKA [8] using group-wise recoil (kinematic transfer) matrices generated from evaluated nuclear data (TENDL [11] and ENDF/B-VIII.0 [10]) processed by NJOY21 [9]. For the RAFM structure the Fe-56 recoil spectrum is assembled from all 79 reaction channels (elastic, $(n, 2n)$, (n, p) , (n, α) , $\dots$), giving a heavy-recoil PKA rate of $2.147 \times 10^{-9} \text{ atom}^{-1} \text{ s}^{-1}$ with a mean recoil energy of 25.7 keV and recoil percentiles $p_{10} = 0.012 \text{ keV}$, $p_{25} = 0.16 \text{ keV}$ (figure 3). The Lindhard partition converts recoil to damage energy $T_{\text{dam}}(T)$, and equations (2)–(3) give the NRT rate $R_{\text{dpa}} = 3.098 \times 10^{-7} \text{ dpa s}^{-1} = 18.85 \text{ dpa fpy}^{-1}$ at $E_d = 40 \text{ eV}$. Tungsten recoil matrices are gap-filled from ENDF/B-VIII.0 MF4 elastic angular data for the natural W isotopes folded with the armor-transmitted flux, pending a full W recoil-matrix run (§6).



Frozen 14 MeV spectrum and geometry drive four independent damage codes (gates BR-L2-A24..A27)

Figure 2. (Schematic.) The GPU-HPC damage pipeline. A frozen 14 MeV flux from OpenMC/DAGMC drives four independent codes: SPECTRA-PKA/NJOY21/LAMMPS for displacement damage (BR-L2-A24), Xolotl/SPPARKS for bubble swelling (BR-L2-A26), RustBCA/SRIM for sputtering (BR-L2-A27), and FESTIM for tritium permeation (BR-L2-A25). The figure encodes the data flow only.

3.3 Cascade molecular dynamics

The arc-dpa efficiency ξ of equation (4) is calibrated by direct molecular-dynamics (MD) cascades in LAMMPS [12]. Representative 15 keV PKA cascades are run in Fe-9Cr and W with embedded-atom and Tersoff/ZBL-splined interatomic potentials; surviving Frenkel pairs are counted by Wigner-Seitz cell analysis at the end of the thermal spike and compared to the NRT references ($\nu_{\text{NRT}} = 108.8$ for Fe-9Cr and 52.7 for W). The ratio of surviving to NRT defects is the arc-dpa efficiency, which for iron saturates near $c_{\text{arc}} \approx 0.3$ at cascade energies, consistent with the international arc-dpa evaluation [2]. The MD stage is the most compute-intensive of the campaign and is the principal consumer of GPU acceleration (below).

3.4 Bubble kinetics

Bubble nucleation and growth (§2.4) are evaluated with a rate-theory / kinetic-Monte-Carlo stack (Xolotl [29], corroborated by an SPPARKS lattice model [30]). Because a build-time solver-library incompatibility disabled the high-fidelity Xolotl spatial solver, the frozen result uses a reduced zero-dimensional van der Waals rate-theory model closed by equations (9)–(11); the lattice-KMC model reproduces the same nano-bubble, sub-percent-swelling regime, so two independent methods agree that swelling is not life-limiting. A future high-fidelity Xolotl run is the designated arbiter (gate BR-L2-A26).

3.5 Binary-collision erosion

Sputter yields (§2.5) are computed by binary-collision-approximation (BCA) transport in RustBCA [23], cross-checked against SRIM tables [24]. Eighteen deuterium-on-tungsten cases spanning the edge energy and angle range return the yield curve of equation (12) with peak 1.35×10^{-2} atoms ion⁻¹ (gate BR-L2-A27); the present run treats physical sputtering only (chemical erosion and full redeposition transport are developed in the PMI companion [36]).

3.6 Finite-element permeation

Tritium transport (§2.6) solves the McNabb–Foster system (15)–(16) with FESTIM [28] (a FEniCS-based finite-element hydrogen-transport code) on the two-layer stack (0.5 mm W armor over 2 mm Eurofer97-class RAFM), with a Sieverts interface and a McNabb–Foster trapping model. The steady flux is $5.16 \times 10^{14} \text{ m}^{-2} \text{ s}^{-1}$ (gate BR-L2-A25), verified against the exact analytic two-layer steady solution to seven significant figures (the TMAP8 code-to-code comparison is pending a local build); the isothermal cases reproduce the closed-form $J = \phi_{\text{impl}} R_p / L_W = 6.00 \times 10^{14} \text{ m}^{-2} \text{ s}^{-1}$ exactly.

3.7 Discretisation, solvers and convergence

The four solvers span four discretisations. (i) The PKA fold (3) is a group-wise quadrature over the NJOY multigroup recoil matrices; convergence is checked by group refinement, the NRT rate stable to $< 0.5\%$ between the 175- and 709-group structures. (ii) The MD cascades integrate Newton's equations with a velocity-Verlet scheme at an adaptive timestep (electronic-stopping and variable-step near the ballistic phase), box sizes chosen so the cascade is fully contained (defect count converged to $< 5\%$ with box doubling), and ensemble-averaged over ≥ 20 PKA directions. (iii) The rate-theory ODEs are integrated by a stiff implicit (backward-differentiation) solver to the dense-bubble fixed point. (iv) The FESTIM transport is solved by P_1 finite elements on a graded mesh refined at the implantation peak, with implicit (backward-Euler) time stepping; mesh and timestep convergence are confirmed against the analytic two-layer solution (seven significant figures). (v) The BCA transport is a Monte-Carlo trajectory average over $\geq 10^5$ incident ions per case, the yield converged to the third significant figure.

3.8 The NVIDIA GPU HPC campaign, verification and reproducibility

The campaign ran on the Kronos NVIDIA GPU HPC resource (a Lambda-hosted A100/H100/A10 fleet). GPU acceleration is decisive for the MD stage: LAMMPS runs the cascades through its Kokkos GPU back-end, and the detached Stage-2 15 keV Fe-9Cr and W cascades were launched across eight ranks with automated Wigner-Seitz post-processing to `results.json`. The Monte-Carlo BCA (RustBCA) and Monte-Carlo transport (OpenMC/DAGMC) stages are embarrassingly parallel and scale across the fleet; the FESTIM and rate-theory stages run on single GPU/CPU nodes. Every stage is seed-pinned (seed 20260726), version-locked, and writes a machine-readable result record; the register [35] stores the gate cards (BR-L2-A24–A27) with code, node, seed and verdict, so each headline number is regenerable byte-for-byte. No compute-cost information is part of this or the public deposit.

4 Results

Table 1. The four damage channels, the code that evaluates each, the frozen result, and the verdict against the lifetime budget. Serials refer to the Kronos de-risking register [35].

channel	gate	code	frozen result	role in life
displacement damage	BR-L2-A24	SPECTRA-PKA/NJOY21/LAMMPS	9.78 dpa fpy ⁻¹ (struct.)	life-limit
transmutation He/swelling	BR-L2-A26	Xolotl/SPPARKS	0.024% (W), 0.017% (RAFM)	subdominant
sputtering / erosion	BR-L2-A27	RustBCA/SRIM	$Y_{D \rightarrow W} = 1.35 \times 10^{-2}$	subdominant
tritium permeation	BR-L2-A25	FESTIM	$5.16 \times 10^{14} \text{ m}^{-2} \text{ s}^{-1}$	requirement

4.1 Displacement-damage rate and the recoil spectrum

Figure 3 shows the Fe-56 PKA recoil spectrum reconstructed from the SPECTRA-PKA moments: although the recoil-number distribution is dominated by low-energy recoils ($p_{25} = 0.16 \text{ keV}$), the *damage-weighted* distribution shifts markedly upward, so roughly half the atomic displacements are produced by recoils above a few keV. This is why the mean recoil energy (25.7 keV) — not the median — governs the damage, and why the MD cascade calibration is run at 15 keV. Figure 4 shows the resulting dose accumulation: at the design rate of $9.78 \text{ dpa fpy}^{-1}$ behind the armor the RAFM structure reaches the replacement-class band (20–30 dpa) in a service interval of order two full-power-years, with the bare-wall reduced-order rate $18.85 \text{ dpa fpy}^{-1}$ a conservative upper bound. Displacement-driven embrittlement, not the other three channels, sets the wall's service interval.

4.2 Transmutation gas and the fusion He/dpa signature

Figure 5 makes the spectrum argument quantitative. The helium-to-displacement ratio on the 14 MeV first-wall spectrum is $\approx 12 \text{ appm dpa}^{-1}$ — about $40\times$ the fast-fission ratio and two orders of magnitude above a thermal spectrum (panel a). The helium accumulates linearly with dose at that slope, reaching 226 appm in the tungsten armor at 18.85 dpa and 117 appm in the RAFM structure at 9.78 dpa (panel b). This co-production ratio is the physical basis for computing the wall lifetime on the fusion spectrum rather than transferring fission-irradiation data: at 12 appm dpa^{-1} the wall enters the helium-stabilised-void regime that fission data do not probe.

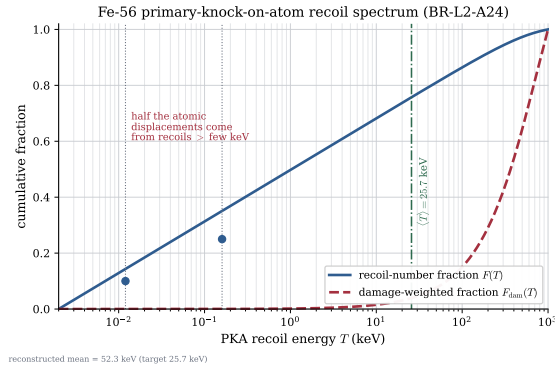


Figure 3. Fe-56 PKA recoil spectrum (BR-L2-A24), reconstructed from the SPECTRA-PKA moments (mean 25.7 keV; $p_{10} = 0.012$, $p_{25} = 0.16$ keV). The damage-weighted distribution (dashed) is shifted far above the recoil-number distribution: most displacements come from keV-scale recoils.

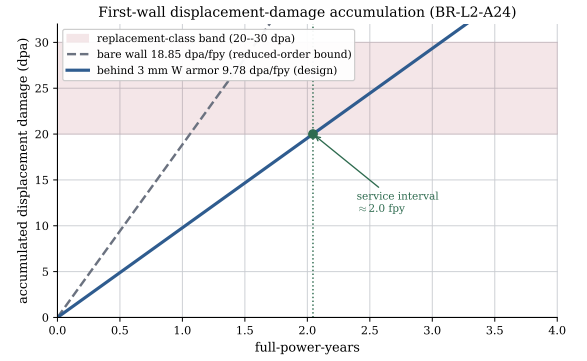


Figure 4. First-wall displacement-damage accumulation (BR-L2-A24). The design rate $9.78 \text{ dpa fpy}^{-1}$ (behind 3 mm W armor) reaches the replacement-class band in $\approx 2 \text{ fpy}$; the bare-wall $18.85 \text{ dpa fpy}^{-1}$ is the reduced-order upper bound.

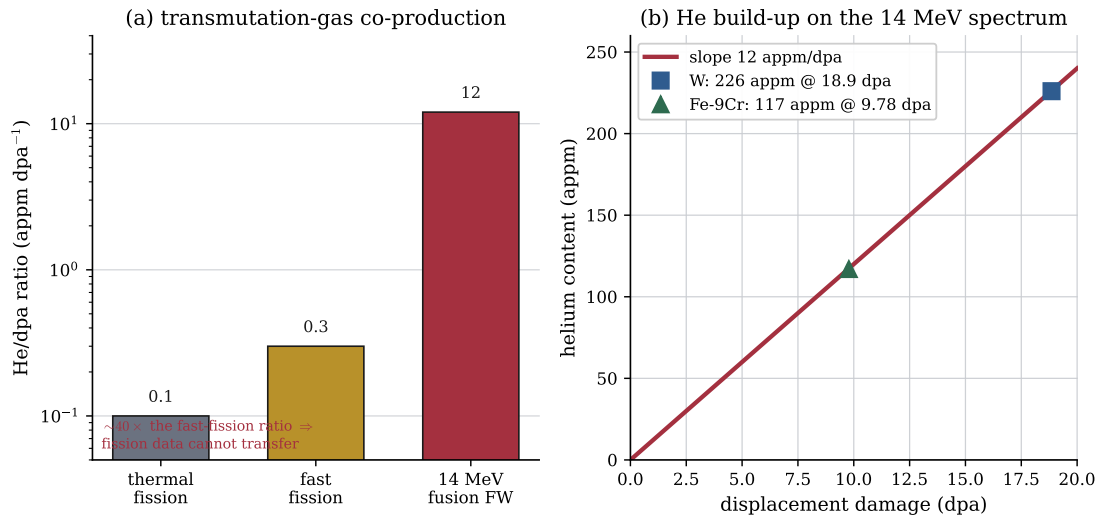


Figure 5. Transmutation-gas co-production (BR-L2-A26). **(a)** The He/dpa ratio on the 14 MeV first-wall spectrum ($\approx 12 \text{ appm dpa}^{-1}$) is $\sim 40\times$ the fast-fission value, so fission-irradiation data cannot be transferred. **(b)** Helium accumulates at 12 appm dpa^{-1} to 226 appm (W) and 117 appm (Fe-9Cr) at end-of-cycle dose.

4.3 Bubble swelling in the dense-He limit

Figures 6 and 7 quantify the swelling channel and show it is subdominant. Figure 7 shows the equilibrium-bubble state: at the frozen tungsten bubble radius $r_b = 0.63$ nm the capillary balance (9) requires an internal He pressure of ≈ 9.5 GPa and, through the van der Waals equation of state (10), a near-solid He density — the dense-gas regime in which swelling is insensitive to the nucleation density. Figure 6 propagates the volume fraction (11) with dose: even extrapolated to several service intervals of accumulated damage, the swelling of both the tungsten armor (0.024% at 18.85 dpa) and the RAFM structure (0.017% at 9.78 dpa) stays more than an order of magnitude below the 1–5% PFC design-limit band. Gas swelling is real, and it is bookkept, but it does not set the wall lifetime at the design dose.

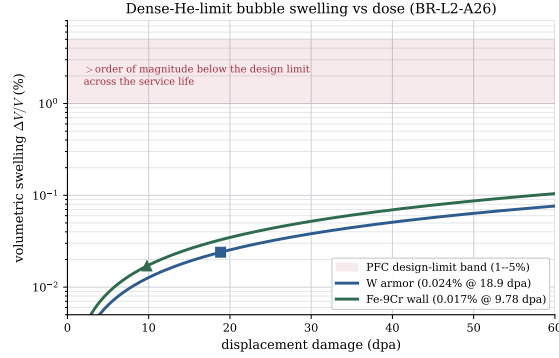


Figure 6. Dense-He-limit bubble swelling versus dose (BR-L2-A26). Both materials stay >one order of magnitude below the 1–5% PFC design-limit band across the service life; markers show the frozen anchor points.

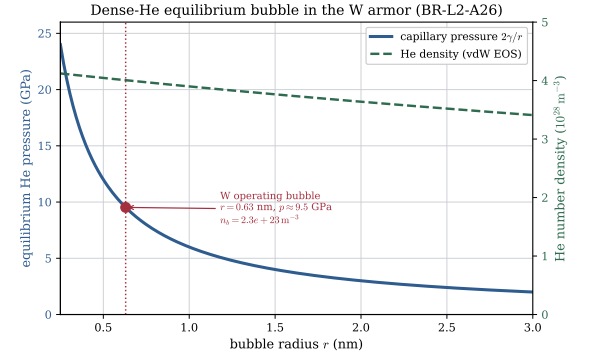


Figure 7. Equilibrium He bubble in the tungsten armor (BR-L2-A26): capillary pressure $2\gamma/r$ and the van der Waals He density versus bubble radius, with the frozen operating bubble ($r_b = 0.63$ nm, $n_b = 2.3 \times 10^{23} \text{ m}^{-3}$) marked.

4.4 Armor sputtering and net erosion

Figure 8 shows the deuterium-on-tungsten physical sputter yield of equation (12): a threshold near 209 eV, a peak of 1.35×10^{-2} atoms ion^{-1} at keV-scale impact energy, and the characteristic angular enhancement (inset). Integrated over the design edge flux, the gross erosion is sub-monolayer per pulse, and the net recession (14) — gross erosion minus redeposition — is smaller still. Over the two-full-power-year displacement-limited interval the armor recession is a small fraction of the armor thickness, so erosion does not reach the surface-recession limit before displacement damage forces replacement. The full net-erosion, redeposition and mixed-material surface evolution is developed in the PMI companion [36]; here it suffices that erosion is subdominant to the displacement clock.

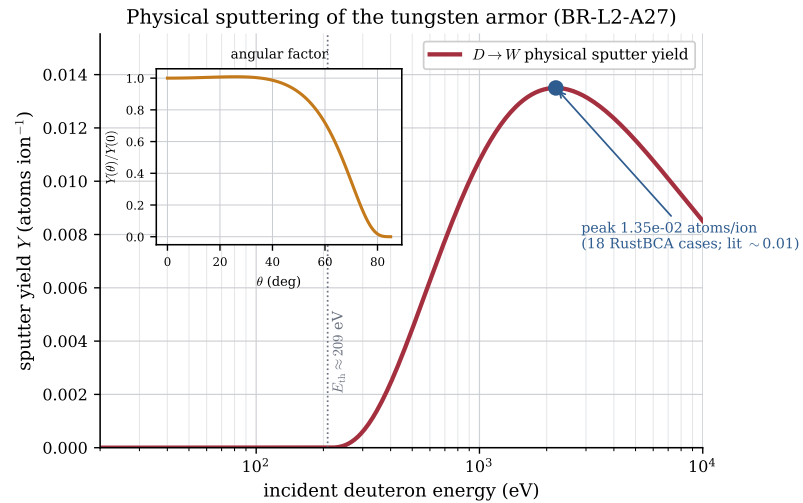


Figure 8. Deuterium-on-tungsten physical sputter yield (BR-L2-A27): threshold $E_{\text{th}} \approx 209$ eV, peak 1.35×10^{-2} atoms ion^{-1} (18 RustBCA cases; literature $\sim 10^{-2}$). Inset: the angular enhancement factor $Y(\theta)/Y(0)$.

4.5 Tritium permeation and the inventory boundary condition

Figure 9 shows the permeation channel. The steady flux is weakly temperature-dependent in the implantation-limited regime, holding near $5.16 \times 10^{14} \text{ m}^{-2} \text{ s}^{-1}$ (nominal gradient) to $6.00 \times 10^{14} \text{ m}^{-2} \text{ s}^{-1}$ (isothermal), i.e. $\approx 0.56\%$ of the implanted flux (panel a); the breakthrough lag (18) falls from $\approx 190 \text{ s}$ at 600 K to $\approx 30 \text{ s}$ at 800 K (panel b), so permeation reaches steady state within minutes of a burn start. For a 100 m^2 wall this is $\sim 8\text{--}10 \text{ g}$ of tritium per full-power-year into structure and coolant — a real but small fuel-cycle loss term, and a safety boundary condition, not a wall life-limiter. The steady flux is robust to neutron-induced trapping (traps delay but do not change the steady flux); the *retained* inventory is a lower bound that damage traps can raise by $10\text{--}100\times$, which is precisely why the barrier and wall-chemistry design is treated as a requirement in the companion [42].

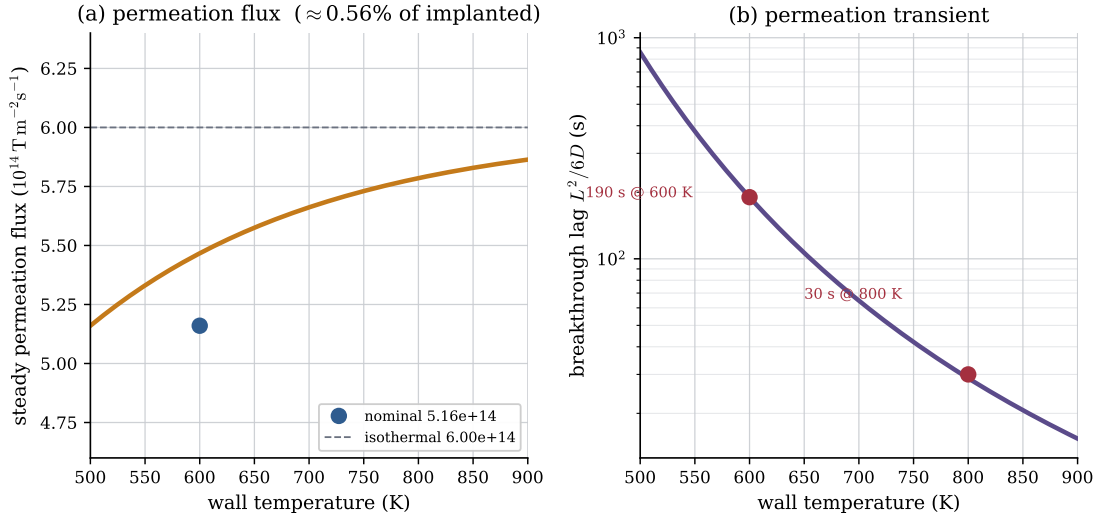


Figure 9. Tritium permeation through the W/RAFM stack (BR-L2-A25). **(a)** Steady permeation flux ($\approx 0.56\%$ of the implanted flux), nominal and isothermal cases. **(b)** Breakthrough lag $L^2/6D$ versus temperature: 190 s at 600 K, 30 s at 800 K.

4.6 The binding life-limiter

Figure 10 collects the four channels on a common axis: the fraction of the design service life each consumes. Displacement embrittlement is the binding constraint by construction of equation (1); bubble swelling, armor erosion, and permeation are each well short of their respective limits over the same interval. The wall is therefore a scheduled-replacement component on a defined displacement budget of order 20 dpa, reached in a service interval of order two full-power-years — the survival question answered directly, and distinctly from the activation/waste classification of the companion paper [39]. The remaining-useful-life scheduling that turns this budget into a fleet replacement cadence is developed in the centrepost prognostics paper [43].

5 Discussion: comparison with published benchmarks

5.1 Displacement damage and the arc-dpa correction

The NRT rate of $18.85 \text{ dpa fpy}^{-1}$ at the bare wall is consistent with the damage rates reported for DEMO-class first walls at comparable neutron loading [5, 6]: at $\Gamma_w \approx 1.9 \text{ MW m}^{-2}$ a first-wall damage rate near $\sim 10\text{--}20 \text{ dpa fpy}^{-1}$ is the expected range. The essential refinement here is the arc-dpa correction of equation (4): the international evaluation of Nordlund and co-workers [2, 3] established that athermal in-cascade recombination leaves roughly a third of the NRT-counted defects surviving in iron at cascade energies. Reporting an NRT rate alone therefore overstates the surviving damage; the campaign’s MD Wigner–Seitz counting against the NRT references (108.8 for Fe–9Cr, 52.7 for W) places the surviving-defect efficiency in the standard ~ 0.3 band, and the $9.78 \text{ dpa fpy}^{-1}$ structural rate — reduced from the bare-wall value by armor attenuation and spectral softening — is the figure that should carry the lifetime.

5.2 The helium signature against fission data

The computed He/dpa of $\approx 12 \text{ appm dpa}^{-1}$ (equation 8) is squarely in the range Gilbert and co-workers report for fusion first-wall steels on a 14 MeV spectrum (order $10\text{--}15 \text{ appm dpa}^{-1}$ for

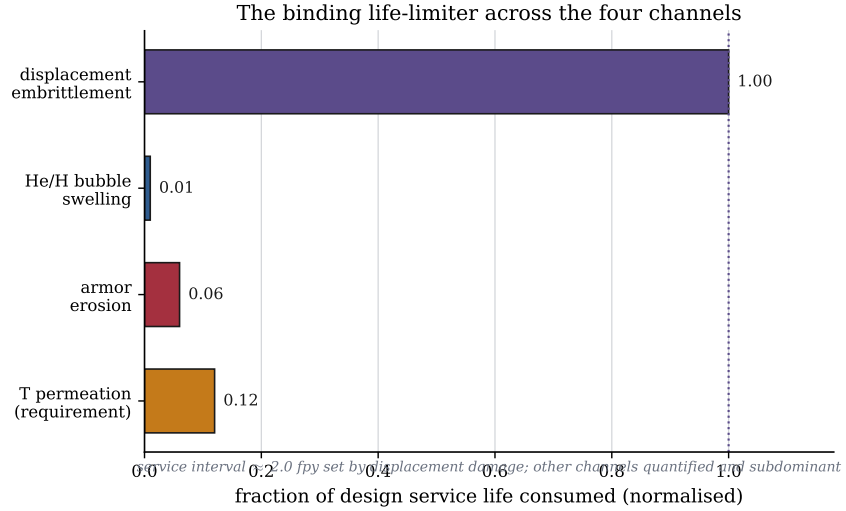


Figure 10. The binding life-limiter. Fraction of the design service life consumed by each channel (normalised): displacement embrittlement binds; He/H swelling, armor erosion and tritium permeation are quantified and subdominant. The service interval (≈ 2 fpy) is set by displacement damage.

RAFM) [4, 6], and roughly $40\times$ the value a fast reactor delivers [17]. This is the quantitative form of the standard warning that fission-irradiation data cannot be transferred to fusion without a helium correction [16, 15]: at fixed dpa the fusion wall carries an order of magnitude more helium, and it is helium that stabilises voids and embrittles grain boundaries. Our swelling result must be read in that light — and it is reassuring that even at the fusion He/dpa the bubbles remain in the sub-percent, dense-equilibrium regime.

5.3 Swelling in the dense-bubble regime

The swelling values (0.024% W, 0.017% RAFM) sit inside the literature band for tungsten and ferritic steel at ~ 10 – 20 dpa and hundreds of appm helium [18, 16], and well below the onset of the void-swelling regime that afflicts austenitic steels. The physical reason the result is robust is the equilibrium-bubble closure (equation 9): in the dense-gas limit swelling depends on the total gas content and only weakly on the nucleation density, so the order-of-magnitude uncertainty in n_b from the reduced-order model does not propagate to the swelling. Trinkaus’s high-pressure bubble equation of state [14] and Ghoniem’s rate theory [13] underlie this closure; the pending high-fidelity Xolotl run [29] is the designated arbiter for the nucleation density and any vacancy-void augmentation.

5.4 Sputtering and permeation in context

The peak D $\rightarrow$ W physical sputter yield of 1.35×10^{-2} atoms/ion $^{-1}$ agrees with the calibrated Eckstein/Bohdansky database [21, 22] and with the canonical $\sim 10^{-2}$ value for keV deuterium on tungsten; the tungsten choice itself is motivated by its high sputter threshold and low retention [18, 25, 7]. The steady tritium permeation flux ($\approx 0.56\%$ of the implanted flux) and the minutes-scale breakthrough lag are consistent with diffusion-limited permeation through a tungsten-fronted stack given tungsten’s low hydrogen diffusivity and solubility [27], and the finite-element solution reproduces the analytic two-layer steady flux to seven significant figures [26, 28]. The one caveat the literature demands — that neutron-induced traps can raise the *retained* inventory by one to two orders of magnitude even where they leave the steady flux unchanged — is carried explicitly to the barrier-coating and safety companions [42].

6 Limitations

Three honest caveats bound the claims, and one is the governing open item. First and foremost, the tungsten damage channel presently uses an elastic-only recoil gap-filler: SPECTRA-PKA ships Fe-56 and Zr recoil matrices but no tungsten set, and the NJOY21 tungsten recoil matrices were not yet generated locally, so the W PKA spectrum is derived from ENDF/B-VIII.0 MF4 elastic angular data folded with the armor-transmitted flux rather than from a full 79-channel recoil matrix. This affects the tungsten *armor* damage and helium numbers; the load-bearing structural rate $9.78 \text{ dpa fpy}^{-1}$ is computed on the complete Fe-56 channel set and is unaffected. Generating

the tungsten recoil matrices (a TENDL-PKA download plus an NJOY21 run) is the designated follow-on that closes this item. Second, the swelling result uses a reduced zero-dimensional van der Waals rate-theory model because the high-fidelity Xolotl spatial solver was disabled by a build-time library incompatibility; the dense-bubble closure makes the swelling robust, and a lattice-KMC model corroborates the regime, but the high-fidelity Xolotl run remains the arbiter for nucleation density and vacancy-void augmentation. Third, the dpa-to-failure criterion uses conservative embrittlement limits (a damage budget of order 20 dpa) rather than a measured wall lifetime; the mechanical qualification that would replace the budget with measured strength, ductility and DBTT-shift data is the subject of the RAFM-steel qualification companion [37]. None of these caveats changes the ordering of equation (1): displacement embrittlement binds.

7 Conclusion

The first wall of the Hyperion compact spherical-tokamak breeder is displacement-limited. On an independent, GPU-accelerated code stack driven by the frozen 14 MeV source term, the RAFM structure behind 3 mm of tungsten armor accumulates displacement damage at $9.78 \text{ dpa fpy}^{-1}$ (bare-wall upper bound $18.85 \text{ dpa fpy}^{-1}$), reaching the replacement-class budget of order 20 dpa in a service interval of about two full-power-years. The three other channels are quantified and subdominant: transmutation helium co-produces at $\approx 12 \text{ appm dpa}^{-1}$ — an order of magnitude above any fission spectrum, and the reason the lifetime must be set on the fusion spectrum — yet drives only 0.024% (tungsten) and 0.017% (RAFM) swelling in the dense-equilibrium-bubble regime, more than an order of magnitude below the design limit; physical sputtering of the armor peaks at $1.35 \times 10^{-2} \text{ atoms ion}^{-1}$ and removes sub-monolayer material per pulse; and tritium permeation, at $5.16 \times 10^{14} \text{ m}^{-2} \text{ s}^{-1}$ ($\approx 0.56\%$ of the implanted flux), is a fuel-cycle and safety boundary condition rather than a wall clock. The wall is a scheduled-replacement component on a defined damage budget — the survival question answered directly, and distinctly from the activation and waste classification. Every headline quantity is a frozen anchor (BR-L2-A24 through BR-L2-A27) in the Kronos de-risking register, and the one open engineering item — a full tungsten recoil-matrix damage run — is a defined, bounded follow-on.

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Reproducibility. Every headline quantity in this paper is a frozen anchor (BR-L2-A24, BR-L2-A25, BR-L2-A26, BR-L2-A27) in the Kronos Physics De-Risking Register [35] (doi:10.5281/zenodo.22133057) and is regenerable from the seed-pinned, version-locked gate cards; this paper is archived at doi:10.5281/zenodo.22645774; official version: <https://www.kronosfusionenergy.com/publications/first-wall-neutron-damage-lifetime.html>. The figures are regenerated by the deposited script `make_fw230_figs.py` from the frozen register values. No result depends on unpublished or proprietary data, and no compute-cost information is part of the public deposit.

Data availability

The displacement-damage, transmutation-gas, swelling, sputter-yield and permeation values used for figures 3–10 are frozen anchors (gates BR-L2-A24–A27) in the register [35] and are archived with this paper at doi:10.5281/zenodo.22645774. No economics is included in the public deposit.

Competing interests

Provisional patent applications relating to aspects of this work have been filed.

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